

E-ASSESSMENT APPLICATIONS AND MANAGEMENT OF EXAMINATION MISCONDUCT AMONG DISTANCE LEARNING STUDENTS IN ENUGU STATE, NIGERIA

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Abstract

This study investigated the relationship between e-assessment applications and management of examination misconduct among distance learning students in Enugu State. Descriptive survey research design was adopted for the study and the population consisted of all the 200 level students of Business Administration in the National Open University of Nigeria (NOUN) Enugu Study Centre. 101 students randomly selected formed the sample for the study. The instruments used were E-Assessment Applications and Examination Misconduct Questionnaire (E-AAEMQ) which was validated and the reliability co-efficient of 0.72 for e-mail, 0.80 for internet browser, 0.88 for file transfer protocol, 0.84 worldwide -web and 0.79 for examination misconduct were obtained. Mean and Standard Deviation were used for answering the research questions while Pearson's Product Moment Correlation (PPMC) was used for testing the hypotheses at .05 level of significance. The result obtained showed that there is a positive significant relationship between e-mail, internet browser, file transfer protocol, world-wide-web and management of examination misconduct by distant learning students in Enugu State. Appropriate recommendations were made based on the findings of the study such as stakeholders must join hands to ensure that this absurdity is nipped in the bud. Equally important is the consideration of outright ban by government on mobile phones into any exam hall. If Industry regulators like the Nigerian Communications Commission (NCC) and academic institutions among others, team-up, the better for the two sectors and the nation at large.

Keyword: Electronic Assessment Applications, Examination Misconduct, Management and Internet browser.

1. INTRODUCTION

Electronic assessment application also known as e-examination is a system that involves the conduct of examinations through the web or the intranet. E-assessment in its broadest sense is the use of information technology for any assessment related activity (Aderogba, 2020). Aderogba K. A. (2020) further said that e-examination reduces the large proportion of workload on examination,

training, grading and reviewing, thus bringing the ability for the institution to release examination results in record time. This is because where the lecturer would spend weeks marking scripts manually, the computer would grade the students as soon as they finish their paper. This innovative strategy according to Okonkwo (2026) was introduced to transform the worrisome examination challenges and to enhance the operational efficiencies. The author

further asserted that NOUN deliberately imbibed the use of technology as best practice for sustainability of the institution and to enhance quality assurance of her products and services to her stakeholders.

The advent of Web applications into the computing technology has brought about a significant revolution in our social life including the traditional system of education and examination. Many institutions are beginning to reevaluate their traditional methods and have considered providing pedagogical materials through the Internet. This trend has also led to so many research works on e-examination as well as Web-based studies, not only in Nigeria, but also globally, such as the works of Yu and Yu (2012). It has to be stated, however, that most of researches focused on the development of Web-based testing and assessment, especially the construction of questions of equal weight and randomization of items on individual's test papers. Ayo, Akinyemi, Adebiye and Ekong (2017) defined electronic assessment as a system that "involves the conduct of examinations through the web or the intranet". Though the definition of Wikipedia is that of electronic assessment (e-assessment), it is related to e-examination. Electronic assessment in its broadest sense is the use of information technology for any assessment related activity. The origin of electronic assessment would naturally be traced to the further deployment of the potentials of the internet and the intranet. Abah, J. A., Honmane, O., Age, T. J., & Ogbule, S. O. (2022) stated that:

"As schools around the world establish connections to the internet and teachers and students gain proficiency with navigating through the vast quantity of readily available information, the true educational potential of the World Wide Web can finally begin to be understood. The web can be a dynamic tool capable of assisting educators in propelling learning exciting and competing levels and of bringing education to any students, anywhere, at any time". (p. 1715)

One of the potentials of the Web is the ability to conduct examination through electronic means. Many scholars have hinted on the advantages of the deployment of e-examination. Awoseyan and Kunle (2010), quoting Prof. OluJegede, the Vice-Chancellor of NOUN, said that:

"Electronic assessment was introduced to address series of anomalies being encountered in the manual tests. He said that the e-examination would remove all human errors recorded in manual examination and create opportunity for students to access their results immediately.... With this, we have removed so many hiccups in the compilation of answer scripts and movement of examination papers from one part of the country to another. The examination is conducted now through the net." ... it would be difficult for students to carry out any form of examination misconduct. (p. 10).

E-mail is a promising instructional and learning tool. However, its strength as an educational tool relies solely on constructing a solid email-based environment and a pedagogically sound message. Email could be used as a feedback, assessment and communication tool. Smith, Whiteley and Smith (2019), over the course of three studies, concluded that email is a “viable alternative means of course delivery assessment”. There is need for better and more frequent use of asynchronous communication in the higher education classroom. They find that asynchronous communication, such as email, can be adopted to effective teaching and assessment. They stress that email can increase faculty-student contact resulting in improved student involvement and motivation.

Furthermore, internet browsing is a kind of orienting strategy. It is supposed to identify something of relevance for the browsing organism. When used about human beings it is a metaphor taken from the animal kingdom. It is used, for example, about people browsing open shelves in libraries or browsing databases or the Internet. In library and information science it is an important subject, both purely theoretically and as applied science aiming at designing interfaces which support browsing activities for the user. Hjørland (2011) provided the following definition: "Browsing is a quick examination of the relevance of a number of objects which may or may not lead to a closer examination or acquisition/selection of (some of) these objects. It is a kind of orienting

strategy that is formed by our “theories,” expectations and subjectivity”.

The World Wide Web can be used to provide instruction and instructional support as well as course assessment. Web-based instruction offers learners unparalleled access to instructional resources, far surpassing the reach of the traditional classroom. It also makes possible learning experiences that are open, flexible, and distributed, providing opportunities for engaging, interactive, and efficient instruction. File Transfer Protocol, commonly referred to as FTP, is a client-server protocol that allows the transfer of entire files. Users typically access files this way through FTP client software. FTP uses Transmission Control Protocol (TCP) to transfer files from one location to another. Sending a small file in an email as an attachment can work in many situations, but for large files this can be a painful process. FTP has made it easier for individuals and businesses to transfer files between different locations. A lighter version of FTP, Trivial File Transfer Protocol (TFTP), is used by some services in order to transfer simpler file types between networks. FTP is considerably larger and more complex than TFTP.

In the country's tertiary institutions, ICTs have resulted in high level of examination misconduct, tagged “e-cheating” among students. “E-cheating” according to Omonijo (2017) is the habit of students engaging ICT devices to indulge in examination misconduct. The menace has now provided a smart way for notorious students to beat the effort of civil

society group and well-meaning Nigerians in curtailing examination fraud. Prior the invention of ICTs, the menace of examination misconduct was a big problem to the nation's education system (Oduwaiye, 2015). The problem is becoming aggravated and unbearable because many examination officials, including University staff, collaborate, connive and collude with students to perpetuate the dysfunctional habit, when students might have bribed them (Olatunbosun, 2009). Examination misconduct discourages candidates from studying hard, denies innocent students the opportunity for admission, delays the processing of examination results, dissatisfies candidates, decreases job efficiency and stunts national growth; it must therefore not be allowed to flourish in Nigeria. Fighting the examination misconduct scourge in Nigeria should be a collective responsibility of all Nigerians.

Several studies have emerged on examination misconduct in Nigeria, but electronic-cheating habit of students as a new form of examination fraud is yet to be given adequate attention in the literature. Omonijo and Nnedum (2018) studied e-cheating habit of students in three selected Universities in Nigeria. Using three selected Universities samples, the study examines methods which students are using engaging ICTs to perpetuate examination misconduct. Relying on raw data of 199 students retrieved, the study attempted to uncover if a significant difference exists in e-cheating habit between ICT-compliant students and other students in rubrics not science-

oriented. It equally tried to discover if there is a significant difference in e-cheating habit between male and female students. Drawing on the recorded data in each institution, the study reported five ICT tools associated with examination misconduct. Using chi-square (X^2) to test the hypotheses, the study reported a significant difference between ICT-compliant students and other students in rubrics not science oriented. Finally, it revealed a significant difference in e-cheating habit between male and female students.

In another study Olawale (2012) conducted a study on E- Exams System for Nigerian Universities with Emphasis on Security and Result Integrity. Based on the analysis of the interviews and study of the existing electronic examination system, some anomalies were discovered and a new e-exams system was developed to eradicate these anomalies. In a similar study, Sanni and Mohammad (2015) conducted a study on computer Based Testing (CBT): An Assessment of Student Perception of JAMB UTME in Nigeria Library and Information Science, Dept. In Nigeria students are assessed using paper and pen on cognitive abilities. The finding revealed among others, that majority of the respondents confirmed that CBT can curb examination malpractice. Majority of candidates were also found to prefer CBT than the conventional way of writing examination. There is therefore need to strategize innovative ways for curbing the menace of examination misconduct in our schools. It is on this background that the researcher investigates on

the relationship between electronic assessment and examination misconduct among distance learning students in Enugu State, Nigeria.

1.1 Statement of the Problem

The value and functionality of any educational system lies in its ability to actualize the goals of education. Till date, examinations still remain the necessary tool for an objective assessment and evaluation of what learners have achieved after a period of schooling. Our youths are no longer studying their books because miracles are happening in their various examination halls. Examination misconduct has spontaneously dominated many of the Nigeria schools including private schools. Exam misconduct is among the serious problems and the worm that is plaguing our educational system nowadays. The examination misconduct act (1999) explains examination misconduct as any act of omission or commission by a person who in anticipation of, before, during or after any examination fraudulently secure any unfair advantage for himself or any other person in such a manner that contravenes the rules and regulations to the extent of undermining the validity, reliability, authenticity of the examination and ultimately the integrity of the certificate issued.

Examination misconduct is any irregular behaviour exhibited by a candidate or anybody charged with the conduct of examination before, during or after the examination that contravenes the rules and regulations governing the conduct of such examination. Though examination misconduct is neither a recent phenomenon nor peculiar to Nigeria or Africa,

the alarming rate of increase in examination misconduct in schools call for concern from all stakeholders in the education sector. The incidence of examination misconduct has become so widespread that there is virtually no examination anywhere at all levels and outside the formal school system without any form of shoddy practice or the other. Examination misconducts are common everywhere and every examination season witnesses the emergence of new and ingenious ways of cheating, stealing of question papers, impersonation, disturbances at examination, obstruction of supervision, forgery of result slip, breach of duty, conspiracy and aiding etc. Furthermore, the introduction of electronic assessment (examination) in the country has revolutionized examination misconduct in the school system. The emergence of technological devices has spawned new and more sophisticated approaches to dishonest conduct during examinations. A lot of academic information is downloaded or stored for direct use in examination hall or for onward transfer via SMS, E-mail etc. to other students anywhere in the country. It is of these premises that the researcher attempt to investigate on the relationship between electronic assessment and examination misconduct among distance learning students in Enugu State, Nigeria

1.2 Purpose of the Study

Generally, this study investigated the relationship between electronic assessment and examination misconduct among distance learning students in Enugu State. Specifically, the objectives of the study are to:

1. assess the relationship between e-mail and management of examination misconduct by distance learning students.
2. examine the relationship between internet browser and management of examination misconduct by distance learning students.
3. determine the relationship between file transfer protocol and management of examination misconduct by distance learning students.
4. ascertain the relationship between world-wide-web and management of examination misconduct by distance learning students.
5. determine the relationship between electronic assessment and examination misconduct among distance learning students in Enugu State

1.3 Research Questions

The study answered the following research questions:

1. What is the extent of e-mail usage in management of examination misconduct among students' of distance learning in Enugu State?
2. What is the extent of internet browser usage in management of examination misconduct among students' of distance learning in Enugu State?
3. What is the extent of file transfer protocol usage in management of examination misconduct among students' of distance learning in Enugu State?

4. What is the extent of World Wide Web usage in management of examination misconduct among students' of distance learning in Enugu State?
5. What is the extent of electronic assessment applications usage in management of examination misconduct among students' of distance learning in Enugu State?

1.4 Hypotheses

The following hypotheses were formulated to guide this study and tested at 0.05 level of significance.

1. There is no significant relationship between e-mail and management of examination misconduct by distance learning students.
2. There is no significant relationship between internet browser and management of examination misconduct by distance learning students.
3. There is no significant relationship between file transfer protocol and management of examination misconduct by distance learning students
4. There is no significant relationship between world-wide-web and management of examination misconduct by distance learning students.
5. There is no significant relationship between electronic assessment applications and examination misconduct among distance learning students in Enugu State

2. METHODOLOGY

The study adopted descriptive survey research design, aimed at providing the opinions of the respondents on electronic assessment and examination misconduct among distant learning students. According to Nworgu (2006), descriptive survey research design aims at collecting data and describing in a systematic way, the characteristics and features of facts about a given population. The design was considered the most suitable in carrying out this study, because it facilitated the collection of data systematically from a sample population. The study was carried out in Enugu State, Nigeria. The population for this study comprised of all the 210 2nd year students of Business Administration in the National Open University of Nigeria (NOUN) Enugu Study Centre. A sample size of 101 students selected through simple random sampling technique which represent 50% of the number of students.

E-Assessment Applications (E-AAQ) and Examination Misconduct Questionnaire (EMQ) were the instruments for the study. E-Assessment Application consisted 16 items (4 items for each variable in for E-Assessment Application) while Examination Misconduct Questionnaire (EMQ) comprised 16 items. Each respondent was required to respond by indicating his/her degree of agreement or disagreement with each statement based on a four point Likert scale of "Strongly Agree (SA), Agree (A), Disagree (D), and Strongly Disagree (SD). Face and content validity of the research instrument was presented to three experts. To establish internal consistency for the E-Assessment Applications (E-AAQ) and

Examination Misconduct Questionnaire (EMQ), the instrument was trial tested and was administered to 20 students outside the study area which were not part of the sample but possessed the same characteristics. Data collected were analyzed using Cronbach's Alpha Statistics which yielded reliability coefficient indices of 0.7 for e-mail, 0.80 for internet browser, 0.88 for file transfer protocol, 0.84 worldwide-web and 0.79 for examination misconduct,. These high reliability coefficients confirmed that these variables were internally consistent.

Prior to the commencement of the study, the researcher visited each of the NOUN Enugu study centres seek for permission before the administration of the E-Assessment Applications (E-AAQ) and Examination Misconduct Questionnaire (EMQ). This was personally done by the researcher with the help of two research assistants. Mean and Standard Deviation was used for answering the research questions while Pearson Product Moment Correlation was also used for testing the hypotheses by comparing it with the critical-r-value. The entire hypotheses formulated were tested at .05 level of significance. In order to take decision as regards the research questions were adjudged based on criterion mean of 2.5 and above indicated the high extent of electronic assessment applications usage in management of examination misconduct among students' of distance learning. In terms of hypotheses, whenever the calculated r-value was greater than the critical r-value, the null hypothesis was rejected and also whenever the

calculated r-value was less than the critical r-value, the null hypothesis was retained.

3. RESULTS AND FINDINGS

3.1 RESEARCH QUESTIONS

- i. What is the extent of e-mail usage in management of examination misconduct among students' of distance learning in Enugu State?

Table 1: Mean and standard deviation on the extent of e-mail usage in management of examination misconduct among students' of distance learning in Enugu State

S/N	Items	$\bar{X}$	SD	Decision
1	I receive an SMS a day before the exams	3.18	0.74	HE
2	Supervision is not with because they teal it electronic exams	3.34	0.55	HE
3	While answering exam question I can download answer from my inbox	3.03	0.67	HE
4	I can e-mail the whole question to my friend to answers and send such the answer	2.16	0.77	LE
5	I can transfer my answer directly into the e-mail box of the exam after proper arrangement	2.78	0.87	HE
	Cluster Mean	2.89	0.72	HE

HE = High Extent, LE = Low Extent

The results in Table 1 show the mean and standard deviation on the extent of e-mail usage in management of examination misconduct among students' of distance learning in Enugu State. The table show that 4 items had their mean of 3.18, 3.34, 3.03 and 2.78 with their standard deviation of 0.74, 0.55, 0.67 and 0.87, this indicated that their mean greater than the criterion mean of 2.50 ($\bar{x} > 2.50$). While 1 items is said be a low extent because its mean of 2.16 with a standard deviation of 0.77 is less than the criterion mean of 2.50 ($\bar{x} > 2.50$) with a grand mean of 2.89 which indicate a high extent of e-mail usage in management of examination misconduct among students' of distance learning in Enugu State.

- ii. What is the extent of internet browser usage in management of examination misconduct among students' of distance learning in Enugu State?

Table 2: Mean and standard deviation on the extent of internet browser usage in management of examination misconduct among students' of distance learning in Enugu State

S/N	Items	$\bar{X}$	SD	Decision
6	While writing I can use my phone to download correct answer	3.08	0.63	HE

7	Examination malpractice is easy through internet browser	3.28	0.63	HE
8	Within internet the world is global village –students can down load the questions before exams time	3.27	0.66	HE
9	Student can pay those in the computer room and send the answer later	2.13	0.82	LE
10	Exams can be taking at during private time using internet explorer	2.87	0.65	HE
Cluster Mean			0.68	HE
		2.93		

HE = High Extent, LE = Low Extent

The results in Table 2 show the mean and standard deviation on the extent of internet browser usage in management of examination misconduct among students' of distance learning in Enugu State. The table show that 4 items had their mean of 3.08, 3.28, 3.27 and 2.87 with their standard deviation of 0.63, 0.63, 0.66 and 0.65, this indicated that their mean greater than the criterion mean of 2.50 ($\bar{x} > 2.50$). While 1 items is said be a low extent because its mean of 2.13 with a standard deviation of 0.82 is less than the criterion mean of 2.50 ($\bar{x} > 2.50$) with a grand mean of 2.93 which indicate a high extent of internet browser usage in management of examination misconduct among students' of distance learning in Enugu State.

- iii. What is the extent of file transfer protocol usage in management of examination misconduct among students' of distance learning in Enugu State?

Table 3: Mean and standard deviation on the extent of file transfer protocol usage in management of examination misconduct among students' of distance learning in Enugu State

S/N	Item	$\bar{X}$	SD	Decision
11	Using the school web page and down load the questions which can be transferred through Firefox	3.25	0.55	HE
12	FTP client can connect anonymously and transfer the files contain exam questions to the students	3.21	0.59	HE
13	File transfer can be done without knowing the user name and pass word hence can aid exams malpractice	2.20	0.91	LE
Cluster Mean		2.88	0.68	HE

HE = High Extent, LE = Low Extent

The results in Table 3 show the mean and standard deviation on the extent of file transfer protocol usage in management of examination misconduct among students' of distance learning in Enugu State. The table show that 2 items had their mean of 3.25 and 3.21 with their standard deviation of 0.55 and 0.59, this indicated that their mean greater than the criterion mean of 2.50 ($\bar{x} > 2.50$). While 1 items is said be a low extent because its mean of 2.20 with a standard deviation of 0.91 is less than the

criterion mean of 2.50 ($\bar{x} > 2.50$) with a grand mean of 2.88 which indicate a high extent of file transfer protocol usage in management of examination misconduct among students' of distance learning in Enugu State.

- iv. What is the extent of World Wide Web usage in management of examination misconduct among students' of distance learning in Enugu State?

Table 4: Mean and standard deviation on the extent of World Wide Web usage in management of examination misconduct among students' of distance learning in Enugu State

S/N	Item	$\bar{X}$	SD	Decision
14	Through world-wide web examination papers can be assess by students	3.15	0.59	HE
15	Through www examination question can be transferred in form of second	3.41	0.39	HE
16	The anonymous www allows you to download files even without engaging into any authentication	2.50	0.51	HE
	Cluster Mean	3.02	0.49	HE

HE = High Extent, LE = Low Extent

The results in Table 4 show the mean and standard deviation on the extent of World Wide Web usage in management of examination misconduct among students' of distance learning in Enugu State. The table show that all the items had their mean of 3.15, 3.41 and 2.50 with their standard deviation of 0.59, 0.39 and 0.51, this indicated that their mean greater than the criterion mean of 2.50 ($\bar{x} > 2.50$) with a grand mean of 3.02 which indicate a high extent of World Wide Web usage in management of examination misconduct among students' of distance learning in Enugu State.

- v. What is the extent of electronic assessment applications usage in management of examination misconduct among students' of distance learning in Enugu State?

Table 5: Mean and standard deviation on the extent of electronic assessment applications usage in management of examination misconduct among students' of distance learning in Enugu State

S/N	Items	$\bar{X}$	SD	Decision
1	I receive an SMS a day before the exams	3.18	0.74	HE
2	Supervision is not with because they teal it electronic exams	3.34	0.55	HE
3	While answering exam question I can download answer from my inbox	3.03	0.67	HE
4	I can e-mail the whole question to my friend to answers and	2.16	0.77	LE

	send such the answer			
5	I can transfer my answer directly into the e-mail box of the exam after proper arrangement	2.78	0.87	HE
6	While writing I can use my phone to download correct answer	3.08	0.63	HE
7	Examination malpractice is easy through internet browser	3.28	0.63	HE
8	Within internet the world is global village –students can download the questions before exams time	3.27	0.66	HE
9	Student can pay those in the computer room and send the answer later	2.13	0.82	LE
10	Exams can be taking at during private time using internet explorer	2.87	0.65	HE
11	Using the school web page and down load the questions which can be transferred through Firefox	3.25	0.55	HE
12	FTP client can connect anonymously and transfer the files contain exam questions to the students	3.21	0.59	HE
13	File transfer can be done without knowing the user name and pass word hence can aid exams malpractice	2.20	0.91	LE
14	Through world-wide web examination papers can be assess by students	3.15	0.59	HE
15	Through www examination question can be transferred in form of second	3.41	0.39	HE
16	The anonymous www allows you to download files even without engaging into any authentication	2.50	0.51	HE
	Cluster Mean	2.92	0.66	HE

HE = High Extent, LE = Low Extent

The data presented on Table 5 revealed the mean and standard deviation of students' responses on the extent of electronic assessment applications usage in management of examination misconduct among students' of distance learning in Enugu State which indicated that 13 items out of 16 items had a mean ranges from 2.60 to 3.41 with corresponding standard deviations ranges from 0.39 to 0.87, while 3 items had their means ranges from 2.13 to 2.20 with corresponding standard deviations ranging from 0.77 to 0.91. These items had their mean greater than the criterion mean of 2.50 ($\bar{X} > 2.50$) with a cluster mean of 2.92 with standard deviations ranging from 0.66, which indicates a high extent of electronic assessment applications usage in management of examination misconduct among students' of distance learning in Enugu State.

3.2 HYPOTHESES

i Hypothesis 1

There is no significant relationship between e-mail and management of examination misconduct by distance learning students.

Table 1: Pearson Product Moment Correlation of relationship between e-mail and management of examination misconduct by distance learning students. (N=101)

Variables	ΣX ΣY	ΣX^2 ΣY^2	ΣXY	r-cal.	r-crit.
E-mail	3548	56550	117851	0.92	0.139
Examination misconduct	8037	264389			

The result presented in Table 1 revealed that the calculated r-value of .92 is greater than the critical r-value of 0.139 at .05 level of significance with 99 degrees of freedom. With this result, the null hypothesis that says there is no significant relationship between e-mail and management of examination misconduct by distant learning students was rejected. This implies that there is a significant relationship between e-mail and management of examination misconduct by distance learning students.

ii Hypothesis 2

There is no significant relationship internet browser and management of examination misconduct by distance learning students.

Table 2: Pearson's Product Moment Correlation of significant relationship between internet browser and management of examination misconduct by distance learning students. (N=101)

Variables	ΣX ΣY	ΣX^2 ΣY^2	ΣXY	r-cal.	r-crit.
Internet browser	3656	60134	121505	0.97	0.139
Examination misconduct	8037	264389			

The result presented in Table 2 revealed that the calculated r-value of .97 is greater than the critical r-value of 0.139 at .05 level of significance with 99 degrees of freedom. With this result, the null hypothesis that says there is no significant relationship between internet browser and management of examination misconduct by distant learning students was rejected. This implies that there is a significant relationship between internet browser and management of examination misconduct by distant learning students.

iii Hypothesis 3

There is no significant relationship between file transfer protocol and management of examination misconduct by distance learning students.

Table 3: Pearson's Product Moment Correlation of significant relationship between file transfer protocol and management of examination misconduct by distant learning students. (N=101)

Variables	ΣX ΣY	ΣX^2 ΣY^2	ΣXY	r-cal.	r-crit.
File transfer protocol	3584	58264	119284	0.91	0.139
Examination misconduct	8037	264389			

The result presented in Table 3 revealed that the calculated r-value of .91 is greater than the critical r-value of 0.139 at .05 level of significance with 99 degrees of freedom. With this result, the null hypothesis that says there is no significant relationship between file transfer protocol and management of examination misconduct by distant learning students was rejected. This implies that there is a significant relationship between file transfer protocol and management of examination misconduct by distant learning students.

iv Hypothesis 4

There is no significant relationship between world-wide web and management of examination misconduct by distant learning students.

Table 4: Pearson's Product Moment Correlation of significant relationship between world-wide web and management of examination misconduct by distance learning students. (N=101)

Variables	ΣX ΣY	ΣX^2 ΣY^2	ΣXY	r-cal.	r-crit.
World-wide web	3584	58264	119284	0.97	0.139
Examination misconduct	8037	264389			

The result presented in Table 4 revealed that the calculated r-value of .97 is greater than the critical r-value of 0.139 at .05 level of significance with 99 degrees of freedom. With this result, the null hypothesis that says there is no significant relationship between world-wide web and management of examination misconduct by distant learning students was rejected. This implies that there is a significant relationship between world-wide web and management of examination misconduct by distant learning students.

4. DISCUSSION OF FINDINGS

The finding of hypothesis one revealed that there is a significant relationship between e-mail and management of examination misconduct by distant learning students. This simply indicates that e-mail platform if effectively manipulated by students can lead to examination malpractice. However, the result is significant because in this digital age most of the exams misconduct are done through e-mail. This finding is supported by the finding of Smith, Whiteley and Smith (2019) who found that email is a "viable alternative means of course delivery assessment.

The finding of hypothesis two also revealed that there is a significant relationship between internet browser and management of examination misconduct by distant learning students. Internet browser is obviously a source of examination malpractice through which candidates post readymade answers which is downloaded free. This finding is supported by the finding of Olawale (2012) who conducted a study on E- Exams System for Nigerian Universities with Emphasis on Security and Result Integrity. Based on the analysis of the interviews and study of the existing electronic examination system, some anomalies were

discovered and a new e-exams system was developed to eradicate these anomalies.

The finding of hypothesis three further revealed that there is a significant relationship between file transfer protocol and management of examination misconduct by distant learning students. In file transfer protocol larger file can be transfer unlike the e-mail environment, therefore that tendency of ultising it for examination misconduct purposes. This finding is supported by the finding of Sanni and Mohammad (2015) who found among others, that majority of the respondents confirmed that CBT can curb examination malpractice.

In the vein hypothesis four revealed that there is a significant relationship between world-wide-web and management of examination misconduct by distant learning students. This finding is true because a lot of bad issues are going on in the whole information technology system which world-wide-web is part of it. This finding is supported by the finding of Omonijoand Nnedum (2012) found asinificant difference between ICT-compliant students and other students in rubrics not scienceoriented.Finally, it revealed a significant difference in e-cheating habit between male and femalestudents.

5. CONCLUSION

Based on the findings of this study, it can be concluded that there is a significant relationship between e-mail, internet browser, file transfer protocol, world-wide-web and management of examination misconduct by distance learning students.

6. RECOMMENDATIONS

Based on the findings of this study, the researchers therefore make the following recommendations:

- i. Stakeholders must join hands to ensure that this absurdity is nipped in the bud, although some maintained that a revolution could only happen in the sector when the Nigerian nation begins to recognize hard-work rather than politicization of the educational system, for the mere reason that most of those in the helm of affairs have their wards schooling abroad.
- ii. Equally important is the consideration of outright ban by government on mobile phones into any exam hall, because some of the sophisticated mobile handsets ‘jammer’ to block reception entirely, although there is need to legislate about this, because installing reception ‘jammer’ could be interpreted as denying one access to information and fundamentals of human rights, which include the right to receive information.
- iii. So, the sooner the industry regulators like the Nigerian Communications Commission (NCC) and academic institutions among others, team-up, the better for the two sectors and the nation at large.
- iv. The Federal and State Ministries of Education being policy makers as regards education, at the federal and state levels in Nigeria should formulate educational policies, bearing in mind,

how to stem examination misconduct which has always led to wrong decisions about the products of schools in Nigeria.

References

- Abah, J. A., Honmane, O., Age, T. J., & Ogbule, S. O. (2022). *Design of single-user-mode computer-based examination system for senior secondary schools in Onitsha North Local Government Area of Anambra State, Nigeria. Village Math Educational Review*, 3(1), 108–134. <https://doi.org/10.5281/zenodo.5820832>
- Aderogba, K. A. (2020). Prospect of Computer Based (CB) And Paper Based (PB) Tests in Adult Education. *Vocational and Technical Education Journal (VOTEJ)* 2(2), 362-372.
- Awoseyan, T. and Kunle, M. (2010). Stress and success of NOUN examinations, *Nigerian Tribune*, pp.10.
- Hjørland, B. (2011). The importance of theories of knowledge: Browsing as an example. *Journal of the American Society for Information Science and Technology*, 62(3), 594– 603.
- Oduwaiye, R. O. (2015). Students' perception of factors and solutions to examination malpractices in Nigeria Universities: A case study of University of Ilorin. *Ilorin Journal of Education*, 24 (7), 23-30.
- Okonkwo, C. A. (2026). *Assessment and students' perceptions of computer self-efficacy and general self-efficacy: A case study of the National Open University of Nigeria. West African Journal of Open and Flexible Learning*, 2(2), 97–132. <https://wajofel.org/index.php/wajofel/article/view/317>
- Olatunbosun, J. B. (2009). Examination malpractice in secondary schools in Nigeria: What Sustains it? *European Journal of Educational Studies* 1(3), 54-64
- Olawale, A. (2012). E- exams system for Nigerian Universities with emphasis on security and result integrity. *International Journal of the Computer, the Internet and Management (IJCIM)*, 18 (2): 1-12.
- Omonijo, D.O. (2017). *Parental influence on wards in escalation of examination misconduct in Nigeria. European Journal of Social Sciences*, 19 (2), 297-307.
- Omonijo, D.O. and Nnedum, O. A. U. (2018). A study of e-cheating habit of students in three selected Universities in Nigeria. *Wulefonia Journal*, 19(9), 387-402.
- Sanni, A. and Mohammad, M. (2015). Computer based testing (CBT): An assessment of student perception of JAMB UTME in Nigeria. *Medicine*, 2: 34-47.
- Smith, C. D., Whiteley, H. E. and Smith, S. (2019). Using e-mail for teaching. *Computers and education* 33:15-25.
- Yu, F. Y., Yu, H. J., (2012). Incorporating e-mail into the learning process: Its impact on student academic achievement and attitudes. *Computers and Education*, 38:117–12.